

Convergence and magic numbers: the optimal value of the all-heads coin game below the fair coin

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Abstract

In the all-heads coin game a player tosses a pool of n coins, each landing heads with probability p , must set aside at least one head each round, and loses on an all-tails round; $w_{n,p}$ denotes the optimal winning probability. We prove that $n \mapsto w_{n,p}$ converges for *every* $p \in (0, 1)$. The proof is a three-line consequence of an exact identity for the increment $w_{n,p} - w_{n-1,p}$, which shows that every decrease is bounded by q^n , $q = 1 - p$. For $p < \frac{1}{2}$ the sequence is nevertheless not eventually monotone: it decreases infinitely often, but the total decrease is summable. This is in sharp contrast to the greedy strategy, whose value oscillates log-periodically without converging; the optimal player cancels that oscillation exactly. The optimal policy has a transparent structure: with j tails the player keeps the smallest *magic number* $\geq j$ — a suffix-maximum record of $w_{n,p}$ — and plays ONE if there is none; we prove that there are infinitely many magic numbers, with $M_{k+1} \leq C(p)M_k$. We derive the exact criterion for the birth of a magic number, a two-phase description of the dynamics between births, and a heuristic law $M_{k+1}/M_k \rightarrow r(p)$ for the growth of magic numbers, confirmed numerically to within a few percent across $0.15 \leq p \leq 0.47$.

1 Introduction

1.1 The game

Fix $p \in (0, 1)$ and write $q := 1 - p$. A player holds n coins, each of which lands heads independently with probability p . In each round all remaining coins are tossed; the player must set aside at least one coin showing heads, and loses if no coin shows heads. The game is won once every coin has been set aside. Writing $w_{n,p}$ for the optimal winning probability, the dynamic-programming principle gives the Bellman equation

$$w_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_{m,p}, \quad w_{0,p} := 1, \quad (1)$$

obtained by conditioning on the number j of tails in the first round: the player then keeps some number $m \in \{j, \dots, n-1\}$ of coins (setting aside $n-m$ of the $n-j$ heads) and continues optimally from state m . Two strategies are natural: ONE sets aside a single head each round (value $a_{n,p}$), and the greedy strategy ALL sets aside *every* head (value $b_{n,p}$):

$$a_{n,p} = p^n + (1 - p^n - q^n) a_{n-1,p}, \quad b_{n,p} = p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j b_{j,p}, \quad a_{0,p} = b_{0,p} := 1. \quad (2)$$

The companion paper [1] determines $w_{n,p}$ exactly at $p = \frac{1}{2}$ (it equals $\frac{1}{2}$), shows that ONE is optimal and $w_{n,p}$ strictly increasing for $p > \frac{1}{2}$, and gives a first-order perturbation expansion for $p < \frac{1}{2}$ near

$\frac{1}{2}$; all of those results are machine-checked in Lean 4. The large- n asymptotics of the greedy strategy are the subject of [2]: for $p < \frac{1}{2}$, $b_{n,p} = \Phi_p(\log n)(1 + O(1/n))$ with Φ_p continuous and $\log(1/q)$ -periodic, and Φ_p is non-constant at first order in $\frac{1}{2} - p$ — the greedy value oscillates and does not converge. A coin-flipping game with a related but distinct scoring rule is studied by van Doorn [4].

The present paper concerns the large- n behaviour of the *optimal* value for $p < \frac{1}{2}$. The natural guess, given the greedy asymptotics, is that $w_{n,p}$ oscillates as well. It does not.

1.2 Main results

Theorem 1.1 (convergence). *For every $p \in (0, 1)$ the limit $W(p) := \lim_{n \rightarrow \infty} w_{n,p}$ exists. Moreover, for every $n \geq 1$,*

$$w_{n-1,p} - w_{n,p} \leq q^n w_{n-1,p} \leq q^n, \quad (3)$$

and $\sum_{n \geq 1} |w_{n,p} - w_{n-1,p}| \leq 1 + q/p$.

Theorem 1.2 (infinitely many decreases). *For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually non-decreasing: $w_{n,p} < w_{n-1,p}$ for infinitely many n .*

A third unconditional result concerns the *records* of the sequence, the states m with $w_m > w_k$ for all $k > m$, which we call magic numbers because they are exactly the states an optimal player reduces the pool to (Proposition 4.3).

Theorem 1.3 (infinitely many magic numbers). *For every $p \in (0, \frac{1}{2})$ there are infinitely many magic numbers $M_1 < M_2 < \dots$, and $M_{k+1} \leq C(p)M_k$ for large k , with $C(p)$ the root $r > 1$ of $(r-1)\log(q/p) = rH(1/r)$ (Theorem 4.2).*

Theorems 1.1 and 1.2 together describe a sequence that converges but is not eventually monotone, with decreases that are individually exponentially small and collectively summable. Both rest on a single exact identity for the increment (Lemma 2.1), whose two error terms have definite signs. The proof of Theorem 1.1 is three lines long once the identity is written down, and is well suited to formal verification.

The contrast with the greedy strategy is striking and, at first sight, paradoxical: $w_{n,p} \geq b_{n,p}$, $b_{n,p}$ oscillates with an amplitude of order 10^{-6} at $p = 0.45$ (and larger for smaller p), yet $w_{n,p}$ converges — the optimal player cancels the greedy oscillation exactly. Section 4 explains how. The Bellman maximum in (1) is attained at a *record* of the value sequence, and the optimal policy is entirely described by the set of records, which we call *magic numbers*: with j tails the player reduces the pool to the smallest magic number $\geq j$ (Proposition 4.3). Between two magic numbers the Bellman equation is linear of ONE type, and the value sits just below the last record, $w_{n,p} \approx w_{M,p} - q^n W(p)$. A new magic number is born exactly when the all-tails probability q^n overtakes the (frozen, extremely small) gap between the last record and the limit; the exact criterion is Proposition 4.6. A two-phase heuristic analysis of this criterion predicts that the magic numbers grow geometrically,

$$M_{k+1}/M_k \longrightarrow r(p),$$

where $r(p) > 1/q$ is the larger root of an explicit transcendental equation (Section 4.2); the prediction agrees with exact computations to within a few percent for $0.15 \leq p \leq 0.47$ (Table 1).

What remains open is the complementary half of non-monotonicity.

Conjecture 1.4. *For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is not eventually non-increasing; equivalently (given Theorem 1.2), it has infinitely many strict local maxima and minima.*

The structural form of the conjecture, which is what the numerics show and what Section 5 aims at, is stronger: there are infinitely many magic numbers $M_1 < M_2 < \dots$, and after each M_k the sequence decreases for a bounded number of steps and then increases until the birth of M_{k+1} (Conjecture 5.6 below). This implies Conjecture 1.4; note that neither statement follows from the other in general (a strictly decreasing tail has infinitely many records).

Numerically this is beyond doubt (Section 6): every decrease is followed, after at most a handful of steps, by a long stretch of increase. Our earlier attempt to prove Conjecture 1.4 by transferring the oscillation of the greedy strategy to w through a renormalisation argument cannot succeed, because it aimed at non-convergence — which Theorem 1.1 rules out. We discuss the status of the conjecture in Section 7.

2 The increment identity and convergence

Throughout, $p \in (0, 1)$ is fixed, $q = 1 - p$, and we abbreviate $w_n := w_{n,p}$, $b_n := b_{n,p}$, $a_n := a_{n,p}$. We write

$$\omega_{n,j} := \binom{n}{j} p^{n-j} q^j \quad (0 \leq j \leq n), \quad K_j^{(n)} := \max_{j \leq m \leq n-1} w_m \quad (1 \leq j \leq n-1),$$

so that $\omega_{n,j} = \mathbb{P}(\text{Bin}(n, q) = j)$ and (1) reads $w_n = \omega_{n,0} + \sum_{j=1}^{n-1} \omega_{n,j} K_j^{(n)}$.

Lemma 2.1 (increment identity). *For every $n \geq 1$,*

$$d_n := w_n - w_{n-1} = p^n (1 - w_{n-1}) - q^n w_{n-1} + \sum_{j=1}^{n-1} \omega_{n,j} (K_j^{(n)} - w_{n-1}), \quad (4)$$

and both the first and the last term on the right are non-negative.

Proof. Since $\sum_{j=0}^n \omega_{n,j} = 1$ we have $w_n = p^n w_{n-1} + q^n w_{n-1} + \sum_{j=1}^{n-1} \omega_{n,j} w_{n-1}$; subtract this from (1). The first term is non-negative because $w_{n-1} \leq 1$, the last because $n-1$ belongs to the index window $\{j, \dots, n-1\}$ of $K_j^{(n)}$, so $K_j^{(n)} \geq w_{n-1}$. $\square$

The identity isolates the only mechanism by which the value can *decrease* when a coin is added: the all-tails event, of probability q^n , which loses the game with n coins but is irrelevant with $n-1$ coins. Everything else — the immediate win, and the freedom to reduce the pool to a better state — helps.

Proof of Theorem 1.1. By Lemma 2.1, $d_n \geq -q^n w_{n-1} \geq -q^n$, which is (3). Hence the negative parts satisfy $\sum_{n \geq 1} d_n^- \leq \sum_{n \geq 1} q^n = q/p$. Since $0 \leq w_n \leq 1$, for every N

$$\sum_{n=1}^N d_n^+ = (w_N - w_0) + \sum_{n=1}^N d_n^- \leq 1 + q/p,$$

so $\sum_n |d_n| \leq 1 + q/p < \infty$ and (w_n) converges. $\square$

Remark 2.2. (a) For $p > \frac{1}{2}$ Theorem 1.1 recovers the convergence proved in [1] by monotonicity, and at $p = \frac{1}{2}$ the sequence is constant. The new content is $p < \frac{1}{2}$. (b) The bound $d_n \geq -q^n w_{n-1}$ is attained at $n = 1$ ($d_1 = p - 1 = -q$) and is asymptotically sharp at every decrease: at the births of magic numbers observed numerically, $-d_n/q^n$ is small but not negligible (e.g. 0.03 at $p = 0.45$, $n = 168$; Section 6). (c) The limit satisfies $a_\infty(p) \leq W(p)$, where $a_\infty(p) = \sum_{k \geq 1} p^k \prod_{j > k} (1 - p^j - q^j) > 0$ is the limit of the ONE value (iterate (2)); in particular $W(p) > 0$. For $p \leq \frac{1}{2}$ one has $w_n \leq w_1 = p$ for all $n \geq 1$: inductively, $w_n \leq p^n + (1 - p^n - q^n)p \leq p$ because $p^n(1 - p) \leq q^n p$ iff $(p/q)^{n-1} \leq 1$. Hence $W(p) \leq p$ for $p \leq \frac{1}{2}$, and $W(p) \rightarrow 0$ as $p \rightarrow 0$.

3 Infinitely many decreases

Proof of Theorem 1.2. Suppose, for contradiction, that $w_n \geq w_{n-1}$ for all $n > M$. Fix $n > M + 1$. For $j \geq M$ the window $\{j, \dots, n-1\}$ lies in the non-decreasing range, so $K_j^{(n)} = w_{n-1}$ and the corresponding terms in (4) vanish. For $j < M$ we use only $0 \leq K_j^{(n)} - w_{n-1} \leq 1$. Hence

$$d_n \leq p^n + \mathbb{P}(\text{Bin}(n, q) \leq M-1) - q^n w_{n-1} \leq p^n + M n^{M-1} p^{n-M+1} - q^n w_M,$$

using $w_{n-1} \geq w_M$ and $\binom{n}{j} \leq n^j$, $q^j \leq 1$ for $j \leq M-1$. As $p < q$ and $w_M > 0$, the right-hand side is $q^n (C n^{M-1} (p/q)^n - w_M) < 0$ for all large n — contradicting $d_n \geq 0$. $\square$

Corollary 3.1. *Let $p \in (0, \frac{1}{2})$. Either (w_n) is eventually non-increasing, or it has infinitely many strict local maxima and infinitely many strict local minima.*

Proof. If (w_n) is not eventually non-increasing, then $d_n > 0$ infinitely often, and by Theorem 1.2 $d_n < 0$ infinitely often. Between a positive and the next negative increment (skipping zero increments) there is a strict local maximum; between a negative and the next positive one a strict local minimum. $\square$

Thus Conjecture 1.4 is exactly what separates the present results from the full statement “infinitely many strict local maxima and minima”.

4 Magic numbers and the structure of the optimal policy

Definition 4.1. A state $m \geq 1$ is a *magic number* (for the fixed p) if $w_m > w_k$ for every $k > m$, i.e. m is a strict suffix-maximum record of the sequence (w_n) . We write $\mathcal{M}_p = \{m_1 < m_2 < \dots\}$ for the set of magic numbers.

By Theorem 1.1, $w_m \geq W(p)$ for every magic number. The set $\mathcal{M}_p$ is infinite, with an explicit bound on the gaps:

Theorem 4.2 (infinitely many magic numbers). *Let $p \in (0, \frac{1}{2})$ and $a_\infty = a_\infty(p) > 0$ the limit of the ONE value. For every $M \geq 1$ put*

$$m_1(M) := \min\{m > M : p^m + \mathbb{P}(\text{Bin}(m, q) \leq M) < a_\infty q^m\}.$$

Then there is a magic number in $(M, m_1(M))$. Consequently $\mathcal{M}_p = \{M_1 < M_2 < \dots\}$ is infinite, $W(p) = \inf_k w_{M_k}$, and $M_{k+1} < m_1(M_k)$, where $m_1(M)/M \rightarrow C(p)$, the root $r > 1$ of $(r-1) \log(q/p) = rH(1/r)$ ($C(0.25) = 2.56$, $C(0.35) = 5.07$, $C(0.45) = 21.1$).

Proof. Fix M and call $m > M$ a *running maximum* if $w_m \geq w_k$ for all $M < k < m$. At a running maximum m , every $K_j^{(m)}$ with $M < j \leq m-1$ is a maximum over a subset of (M, m) , hence $K_j^{(m)} \leq w_m$; and $K_j^{(m)} \leq 1$ for $j \leq M$. The Bellman equation therefore gives

$$w_m \leq p^m + \mathbb{P}(\text{Bin}(m, q) \leq M) + w_m(1 - q^m - \mathbb{P}(\text{Bin}(m, q) \leq M)),$$

i.e. $w_m (q^m + \mathbb{P}(\text{Bin}(m, q) \leq M)) \leq p^m + \mathbb{P}(\text{Bin}(m, q) \leq M)$. Since $w_m \geq a_\infty$, this is impossible for $m \geq m_1(M)$ (note that $\mathbb{P}(\text{Bin}(m, q) \leq M) \leq (M+1)m^M p^{m-M}$ is eventually much smaller than q^m because $p < q$, so $m_1(M) < \infty$). Hence all running maxima lie in $(M, m_1(M))$. If $\sup_{k>M} w_k$ were not attained, there would be running maxima at arbitrarily large indices; so it is attained, and its largest maximiser m^* satisfies $w_{m^*} > w_k$ for all $k > m^*$: m^* is a magic number, and $m^* < m_1(M)$. The asymptotics of m_1 is the large-deviation estimate $\mathbb{P}(\text{Bin}(rM, q) \leq M) = \exp(-rM \text{KL}(1/r \| q) + o(M))$, and $\text{KL}(1/r \| q) > \log(1/q)$ iff $(r-1) \log(q/p) > rH(1/r)$. $\square$

The bound is not sharp — on the exact trajectories M_{k+1}/M_k is 1.5–1.6 against $m_1(M_k)/M_k = 2.6$ for $p = 0.25$, 2.4–2.9 against 5.0 for $p = 0.35$, and 11.1 against 20.5 for $p = 0.45$ (the ratio law of Section 4.2 predicts the true values) — but it is unconditional and uniform in p . Note what the argument does *not* give: it does not exclude an eventually non-increasing tail, along which every index is a record; that is Conjecture 1.4.

The relevance of magic numbers is that they are precisely the states the optimal player reduces to.

Proposition 4.3 (optimal policy). *For $n \geq 2$ let $R_n := \{m \in [1, n-1] : w_m > w_k \text{ for all } m < k \leq n-1\}$ be the set of records of w seen from the horizon $n-1$; note $n-1 \in R_n$. Then for $1 \leq j \leq n-1$ the maximum in (1) is attained at $m = \min\{m \in R_n : m \geq j\}$. A state belongs to R_n for every $n > m$ if and only if it is a magic number. In particular, if M is the largest magic number below n and w is non-decreasing on $[M+1, n-1]$, then $R_n = (\mathcal{M}_p \cap [1, M]) \cup \{n-1\}$, and the rule reads: with j tails, keep the smallest magic number $\geq j$; if there is none below n , set aside a single head (strategy ONE).*

Proof. Let m^* be the largest index in $[j, n-1]$ at which $\max_{j \leq m \leq n-1} w_m$ is attained. Then $w_{m^*} > w_k$ for $m^* < k \leq n-1$, so $m^* \in R_n$. If some $m \in R_n$ satisfied $j \leq m < m^*$, then $w_m > w_{m^*}$, contradicting maximality; hence $m^* = \min\{m \in R_n : m \geq j\}$. The characterisation of magic numbers is Definition 4.1. For the last claim, let $m \in R_n$ with $m \leq M$. Then $w_m > w_k$ for $m < k \leq n-1$, in particular $w_m > w_M$ if $m < M$, and $w_M > w_k$ for all $k > M$ since M is magic; so $w_m > w_k$ for all $k > m$ and m is magic. Conversely magic numbers $\leq M$ lie in R_n . Finally, $M < m < n-1$ is impossible for $m \in R_n$ because w is non-decreasing on $[M+1, n-1]$. $\square$

Remark 4.4. The proposition explains the cancellation of the greedy oscillation. The greedy player keeps *all* j tails; his value is a fixed linear functional of the initial data and inherits the log-periodic profile of the binomial smoothing kernel. The optimal player instead aims at a fixed target below n (the last magic number M) whenever $j \leq M$, which happens with probability $\mathbb{P}(\text{Bin}(n, q) \leq M) \approx 1$ for $n < M/q$. In this regime the value is pinned to w_M up to the all-tails probability, and the binomial smoothing that drives the greedy oscillation is switched off.

Example 4.5 (what the optimal player actually does). The rule of Proposition 4.3 produces behaviour that looks paradoxical until one knows the records. Take $p = 0.45$ and $n = 8$ coins. The values are $w_4 = 0.41118 > w_7 = 0.41112 > w_5 = 0.41108 > w_6 = 0.41104$ (and $w_1 > w_2 > w_3 > w_4$), so the records seen from the horizon 7 are $R_8 = \{1, 2, 3, 4, 7\}$: the states 5 and 6 are never worth keeping, because 7 coins are better than either. The optimal move after the first toss is therefore

heads k	1	2	3	4	5	6	7
coins kept	7	7	7	4	3	2	1
heads set aside	1	1	1	4	5	6	7
heads re-tossed	0	1	2	0	0	0	0

With three heads the player sets aside a single one and deliberately throws the other two back into the pool; with four heads he sets aside all four. The number of heads set aside is monotone in k (it equals $n - m^*(n - k)$ with m^* non-decreasing, so this is a theorem), but the number of heads thrown back is not. This is the smallest such example for $p = 0.45$: for $n \leq 7$ the optimal move is always ALL or forced, because w is decreasing on $\{1, \dots, 6\}$ seen from any horizon ≤ 6 . (For p closer to $\frac{1}{2}$, where $w_6 > w_4$, the same phenomenon appears at $n = 7$; for $n \leq 6$ it cannot occur for any $p < \frac{1}{2}$ near $\frac{1}{2}$, since w decreases on $\{1, \dots, 5\}$ by Corollary 4.11 of [1].)

The effect grows with n . For $n = 20$ and $p = 0.45$ the records below 20 are 1, 2, 3, 4, 15, and the player keeps 19 coins for $k \leq 4$ heads, exactly 15 coins for $5 \leq k \leq 15$ heads, and sets aside everything for $k \geq 16$: with fifteen heads he throws ten of them back, with sixteen none. Any pool of size between 5 and 19 is worth less than a pool of exactly 15, and the player steers to 15 whenever the tails allow it.

4.1 Between two magic numbers: the One-type recursion

Let M be the largest magic number below n , and suppose w is non-decreasing on $[M + 1, n - 1]$ (this is the situation after the short transient following a birth, see Section 6). By Proposition 4.3, $K_j^{(n)} = w_{n-1}$ for $j > M$, and $K_j^{(n)} = w_{m(j)}$ for $j \leq M$, with $m(j)$ the smallest magic number $\geq j$. Grouping the sum in (4) by magic numbers $m_1 < \dots < m_K = M$ and writing $\gamma_k(n) := w_{m_k} - w_{n-1} \geq 0$, the identity becomes

$$d_n = p^n(1 - w_{n-1}) - q^n w_{n-1} + \sum_{k=1}^K \mathbb{P}(m_{k-1} < \text{Bin}(n, q) \leq m_k) \gamma_k(n), \quad m_0 := 0. \quad (5)$$

Equivalently, $w_n = A_n + (1 - B_n)w_{n-1}$ with $A_n = p^n + \sum_k \mathbb{P}(m_{k-1} < \text{Bin}(n, q) \leq m_k) w_{m_k}$ and $B_n = p^n + q^n + \mathbb{P}(\text{Bin}(n, q) \leq M)$: a linear recursion of exactly the type analysed in [1, §3.1 and §4].

Proposition 4.6 (birth criterion). *In the situation above, $n - 1$ is a magic number (equivalently $w_n < w_{n-1}$, and then $w_k < w_{n-1}$ for all $k \geq n$ until the next increase past w_{n-1}) if and only if*

$$q^n w_{n-1} > p^n(1 - w_{n-1}) + \sum_{k=1}^K \mathbb{P}(m_{k-1} < \text{Bin}(n, q) \leq m_k) \gamma_k(n). \quad (6)$$

Proof. This is (5) with $d_n < 0$. □

The criterion is exact but implicit, since the gaps $\gamma_k(n)$ depend on the trajectory. Its dynamics are nevertheless transparent, because the last record dominates. Ignoring the older records, the gap $\gamma(n) := \gamma_K(n) = w_M - w_{n-1}$ evolves by $\gamma(n+1) = \gamma(n) - d_n$, i.e.

$$\gamma(n+1) = \mathbb{P}(\text{Bin}(n, q) > M) \gamma(n) + q^n w_{n-1} - p^n(1 - w_{n-1}), \quad (7)$$

a one-dimensional recursion with an explicit binomial coefficient. It exhibits two phases.

Phase 1, $M < n \lesssim M/q$. The coefficient $\mathbb{P}(\text{Bin}(n, q) > M)$ is exponentially small (the mean nq of the binomial is below M), so $\gamma(n) \approx q^n w$: the value sits just below the record, $w_n \approx w_M - q^n W(p)$, and the increments $d_n \approx q^n(1 - q)W(p)$ track q^n exactly. This is visible as the straight segments following each vertical line in Figure 1.

Freeze, $n \approx M/q$. Once nq passes M , the coefficient $\mathbb{P}(\text{Bin}(n, q) > M)$ tends to 1 and the gap stops shrinking: $\gamma(n) \rightarrow \gamma_M(\infty) = w_M - W(p)$, which is therefore of order $q^{M/q}$. (Observed: $p = 0.45$, $M = 15$: $\gamma = 7.8 \cdot 10^{-9}$ against $q^{27} = 1.0 \cdot 10^{-7}$; $p = 0.42$, $M = 43$: $1.9 \cdot 10^{-19}$ against $q^{74} = 3 \cdot 10^{-18}$.)

Phase 2, $n \gtrsim M/q$. Now $d_n \approx \mathbb{P}(\text{Bin}(n, q) \leq M) \gamma_M(\infty) - q^n W(p)$, with $\mathbb{P}(\text{Bin}(n, q) \leq M) \approx \binom{n}{M} p^{n-M} q^M$. The increments leave the q^n line and decay more slowly; the next magic number is born at the first n with

$$\binom{n}{M} \left(\frac{p}{q}\right)^{n-M} \gamma_M(\infty) < W(p). \quad (8)$$

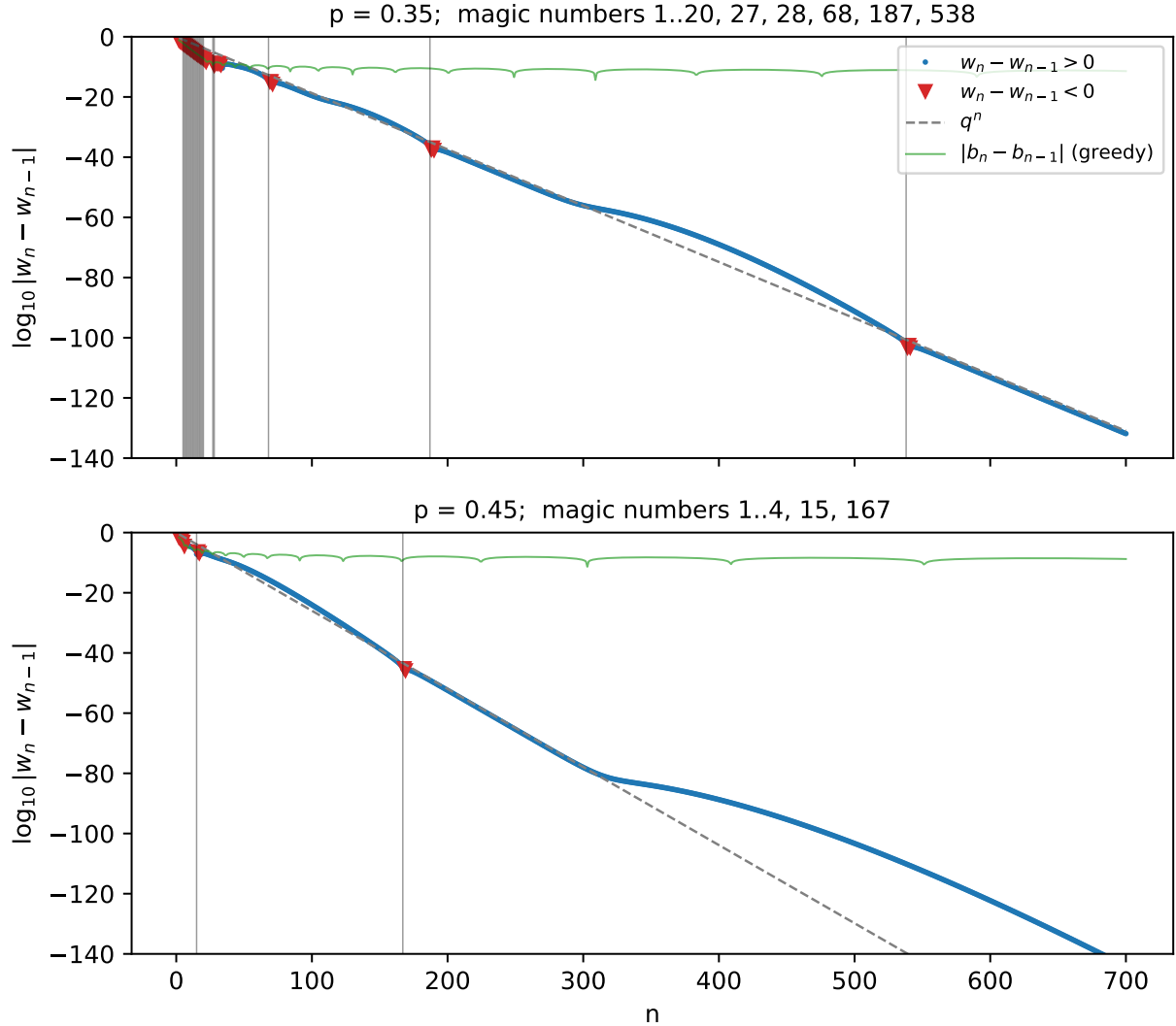


Figure 1: Increments $|w_n - w_{n-1}|$ on a logarithmic scale for $p = 0.35$ (top) and $p = 0.45$ (bottom), $n \leq 700$, computed from the exact Bellman recursion in 220-digit arithmetic. Red triangles mark decreases, vertical lines the magic numbers, the dashed line is q^n , and the green curve is the greedy increment $|b_n - b_{n-1}|$, which does not decay. After each birth the increments follow q^n (Phase 1), leave that line at $n \approx M/q$ (Phase 2), and the next birth occurs when they cross it again.

4.2 The spacing of magic numbers

With Theorem 4.2 the central quantitative question about the optimal value below the fair coin becomes: *how are the magic numbers spaced, as a function of p ?* We collect what is proved, what is conjectured, and the numerical evidence.

Upper bound (proved). $M_{k+1} < m_1(M_k)$ with $m_1(M)/M \rightarrow C(p)$, the root $r > 1$ of $(r-1)\log(q/p) = rH(1/r)$, $H(x) = -x\log x - (1-x)\log(1-x)$ (Theorem 4.2).

Lower bound (conditional). At a birth after a complete recovery, the older-record term must decay faster than q^n (Proposition 5.5 below), which forces $M_{k+1} + 1 \geq q(M_k - K)/(1-2p)$ provided the block $(M_k - K, M_k]$ dominates that term ((12)). Note $q/(1-2p) > 1/q$ always, since $1-2p = 2q-1$ and $q^2 - (2q-1) = p^2 > 0$: a birth can only occur after the freeze at M_k/q .

The law (conjectured). Insert $\log \gamma_M(\infty) \approx (M/q)\log q$ and $\log \binom{rM}{M} \approx rM H(1/r)$ into the birth criterion (8); dividing by M the criterion becomes $g_p(r) > 0$ with

$$g_p(r) := (r-1)\log \frac{q}{p} - r H(1/r) + \frac{1}{q} \log \frac{1}{q}. \quad (9)$$

A short computation shows $g_p(1/q) = 0$ identically — the freeze point is exactly marginal — and g_p is negative just above $1/q$ and eventually positive.

Conjecture 4.7 (ratio law). *For $p \in (0, \frac{1}{2})$ let $r(p) > 1/q$ be the larger root of $g_p(r) = 0$. Then $M_{k+1}/M_k \rightarrow r(p)$ as $k \rightarrow \infty$.*

The same number arises from the exact recursion of Section 5.1 without the freeze heuristic: there the birth is at $n_b = x_b n^*$, $n^* = qM/(1-2p)$, with $J(p, x_b) = I(p)$ in (14), and $x_b q/(1-2p)$ coincides with the root of g_p to all computed digits for every p in Table 1 (the two equations are the same after the substitution $x = rM$). The function $r(p)$ increases from $r(0^+) = 1$ to $r(\frac{1}{2}^-) = \infty$; near $p = \frac{1}{2}$, with $\varepsilon := \log(q/p) \rightarrow 0$, the equation reads $(r-1)\varepsilon \approx \log r - (2\log 2 - 1)$, so $r(p) \sim \varepsilon^{-1} \log(1/\varepsilon)$. Always $q/(1-2p) < r(p) < C(p)$.

p	$q/(1-2p)$	$r(p)$	$C(p)$	observed M_{k+1}/M_k (increasing k)
0.15	1.214	1.255	1.62	1.208, 1.209, 1.212, 1.215, 1.218, 1.221, 1.224, 1.227, 1.230
0.20	1.333	1.426	2.00	1.331, 1.345, 1.355, 1.362, 1.372, 1.382, 1.388, 1.395
0.25	1.5	1.696	2.56	1.49, 1.56, 1.59, 1.61
0.30	1.75	2.155	3.45	1.889, 1.971, 2.030, 2.071, 2.099
0.35	2.167	3.040	5.07	2.43, 2.75, 2.88
0.40	3.0	5.155	8.63	4.33, 4.85
0.42	3.625	6.970	11.5	6.35
0.45	5.5	13.25	21.0	11.1
0.47	8.83	26.46	39.9	24.3
0.49	25.5	110.5	151	($M_1 = 152$)

Table 1: Spacing of magic numbers: the conditional lower bound $q/(1-2p)$, the conjectured limit $r(p)$, the proved upper bound $C(p)$, and the observed ratios of consecutive isolated magic numbers on the exact trajectories ($n \leq 2000$ for $p \leq 0.4$, $n \leq 700$ otherwise). The observed ratios increase with k towards $r(p)$; for $p = 0.15$ and 0.20 they cross the lower bound $q/(1-2p)$ around the fourth cycle and keep rising, so the slow convergence at small p is not a change of law.

The two limits. As $p \uparrow \frac{1}{2}$ the first non-trivial magic number escapes to infinity, $M_1(p) \approx \frac{3}{2}/(\frac{1}{2}-p)$ (152 at $p = 0.49$, 27 at 0.47, 15 at 0.45), and the ratio $r(p) \rightarrow \infty$: on any fixed range of n the sequence looks like the $p > \frac{1}{2}$ case (decreasing on $\{1, \dots, 5\}$, then increasing), which has no magic numbers at all. This is why the first-order theory of [1], which works at fixed n as $\frac{1}{2} - p \rightarrow 0$, sees no local maximum: the first step of the staircase sits at $n \sim (\frac{1}{2} - p)^{-1}$, beyond every order of the expansion. As $p \downarrow 0$, $r(p) \rightarrow 1$ and the initial stretch on which greedy is optimal and every n is a record grows without bound ($n_0(p) = 69, 133, 286, 690$ for $p = 0.25, 0.2, 0.15, 0.1$).

Relation to Conjecture 1.4. Theorem 4.2 leaves two possibilities for the long-run behaviour: either the records are eventually isolated and geometrically spaced, with w increasing between the runs of decreases that follow each birth — the *staircase* of Example 4.5, in which the optimal player plays ONE except for one ALL move per cycle at $n \approx M_k/q$ that lands exactly on M_k — or from some point on every index is a record and w is non-increasing forever, in which case the optimal policy is ALL. Conjecture 1.4 says the first alternative holds for every $p < \frac{1}{2}$; Theorem 5.9 below excludes the second unless the tail starts within a factor $q^2/(1 - 2p)$ of the previous record.

5 Rigorous elements of the cycle theory

This section isolates what can be proved about one cycle directly from the increment identity (4). Throughout, $M' < M$ are consecutive magic numbers, and for $n > M$ we write

$$\gamma(n) := w_M - w_{n-1} (\geq 0), \quad x_j := \max_{j \leq m \leq M} w_m \quad (1 \leq j \leq M), \quad s_n := \sum_{j=1}^{M'} \omega_{n,j} (x_j - w_{n-1}) (\geq 0),$$

$$U_n := \mathbb{P}(\text{Bin}(n, q) \leq M'), \quad V_n := \mathbb{P}(\text{Bin}(n, q) > M),$$

so that $\gamma(M+1) = 0$, s_n is the *older-record term*, and U_n, V_n are exponentially small for $M'/q \ll n \ll M/q$ (the mean nq of the binomial lies between M' and M).

Lemma 5.1 (exact cycle identity). *For every $n > M$ there is $\theta_n \in [0, 1]$ such that*

$$\gamma(n+1) = (U_n + V_n - \theta_n(V_n - q^n))\gamma(n) + q^n w_{n-1} - p^n(1 - w_{n-1}) - s_n. \quad (10)$$

Moreover $\theta_n = 0$ whenever $w_{n-1} \geq w_j$ for all $M < j < n$ (in particular whenever w is non-decreasing on $[M+1, n-1]$), and $\theta_n \leq 1$ always. Consequently, for every $n > M$,

$$(U_n + q^n)\gamma(n) \leq \gamma(n+1) - [q^n w_{n-1} - p^n(1 - w_{n-1}) - s_n] \leq (U_n + V_n)\gamma(n). \quad (11)$$

Proof. Split the sum in (4) according to $j \leq M'$, $M' < j \leq M$, $M < j \leq n-1$. For $j \leq M'$, $K_j^{(n)} = x_j$ since $w_{M'} > w_k$ for all $k > M'$; this gives s_n . For $M' < j \leq M$, $K_j^{(n)} = w_M$ (every value in (M', M) is below w_M , otherwise its position would be a record between M' and M , and every value after M is below w_M); these terms contribute $\mathbb{P}(M' < \text{Bin}(n, q) \leq M) \gamma(n) = (1 - U_n - V_n)\gamma(n)$. For $M < j \leq n-1$ the terms $Q_n := \sum_{j=M+1}^{n-1} \omega_{n,j} (K_j^{(n)} - w_{n-1})$ satisfy $0 \leq Q_n \leq (V_n - q^n)\gamma(n)$, because $w_{n-1} \leq K_j^{(n)} \leq w_M$ and the weights sum to $\mathbb{P}(M < \text{Bin}(n, q) \leq n-1) = V_n - q^n$; put $\theta_n := Q_n/((V_n - q^n)\gamma(n))$. Then $\gamma(n+1) = \gamma(n) - d_n$ is (10). If $w_{n-1} \geq w_j$ on (M, n) then $K_j^{(n)} = w_{n-1}$ for $j > M$ and $Q_n = 0$. $\square$

Lemma 5.1 says that, up to the exponentially small coefficient of $\gamma(n)$, the gap to the last record is *memoryless*: $\gamma(n+1) \approx q^n w_{n-1} - s_n$, the all-tails term minus the older-record term. This is the rigorous form of Phase 1. Three consequences are immediate.

Corollary 5.2 (dip criterion). *For $n > M$: if $q^n w_{n-1} - p^n(1 - w_{n-1}) - s_n > (1 - U_n - q^n)\gamma(n)$ then $w_n < w_{n-1}$; if $w_n < w_{n-1}$ then $q^n w_{n-1} - p^n(1 - w_{n-1}) - s_n > (1 - U_n - V_n)\gamma(n)$. In particular the first dip after M has size $\gamma(M+2) = q^{M+1}w_M - p^{M+1}(1 - w_M) - s_{M+1}$ exactly, and defining the marginality $m := s_{M+1}/(q^{M+1}w_M) \in (0, 1)$, the run of dips following the birth continues at n as long as $q^n w_{n-1} - s_n > (1 - \varepsilon_n)(q^{n-1}w_{n-2} - s_{n-1})$ with $\varepsilon_n \leq U_n + V_n$, i.e. essentially as long as $s_{n-1} - s_n > (1 - q)q^{n-1}w$: the older-record term must drop by more than the all-tails term does.*

Proof. $w_n < w_{n-1}$ iff $\gamma(n+1) > \gamma(n)$; apply (11). $\square$

Lemma 5.3 (older-record term). *For $n > M$, with $\rho_n^+ := p(n+1)/(n+1-M')$ and $U'_n := U_n - p^n$,*

$$p(s_n - d_n U'_n) \leq s_{n+1} \leq \rho_n^+(s_n - d_n U'_n).$$

In particular $s_{n+1} \leq \rho_n^+ s_n$ whenever $d_n \geq 0$.

Proof. $\omega_{n+1,j} = \omega_{n,j} p(n+1)/(n+1-j)$ and $p \leq p(n+1)/(n+1-j) \leq \rho_n^+$ for $1 \leq j \leq M'$; and $x_j - w_n = (x_j - w_{n-1}) - d_n$, with $\sum_{j \leq M'} \omega_{n,j} = U'_n$. $\square$

Proposition 5.4 (freeze bound). *Let $n_1 := \max\{n > M : U_k + V_k \leq \frac{1}{2} \text{ for } M < k \leq n\}$. Then $\gamma(n) \leq C_q q^{n-1}$ for $M < n \leq n_1 + 1$, $C_q := 2q/(2q-1)$, and*

$$0 \leq w_M - W(p) \leq C_q q^{n_1} + q^{n_1+1}/p.$$

Since $V_n \leq \frac{1}{4}$ for $n \leq (M - c\sqrt{M \log M})/q$ and $U_n \leq \frac{1}{4}$ for $n \geq M$ (Chernoff), $n_1 \geq (M - c\sqrt{M \log M})/q$ and hence $w_M - W(p) \leq C q^{(M - c\sqrt{M \log M})/q}$.

Proof. By (11), $\gamma(n+1) \leq q^n + \frac{1}{2}\gamma(n)$ for $M < n \leq n_1$; induction from $\gamma(M+1) = 0$ gives $\gamma(n) \leq C_q q^{n-1}$ (the constant satisfies $1 + C_q/(2q) = C_q$). Finally $w_{n-1} - W(p) \leq \sum_{i \geq n} d_i^- \leq q^n/p$ by Theorem 1.1, so $w_M - W(p) \leq \gamma(n) + q^n/p$. $\square$

Proposition 5.5 (the older-record term must decay faster than q at a birth). *Suppose $w_{M-1} \geq w_j$ for all $M' < j < M$ (complete recovery before the birth). Then*

$$s_M \geq q^M w_{M-1} - p^M(1 - w_{M-1}), \quad s_{M+1} < q^{M+1} w_M - p^{M+1}(1 - w_M),$$

hence

$$\frac{s_{M+1}}{s_M} < q \cdot \frac{w_M}{w_{M-1}} \cdot \left(1 - \frac{(p/q)^M(1 - w_{M-1})}{w_{M-1}}\right)^{-1} = q(1 + O(q^{M'}/p + (p/q)^M)),$$

using $0 \leq d_M \leq \sum_{i > M'} d_i^- \leq q^{M'+1}/p$.

Proof. $M-1$ is not a record and M is, so $d_M \geq 0$ and $d_{M+1} < 0$. Evaluate (4) at $n = M$ (where, by the hypothesis, the terms with $M' < j < M$ vanish and those with $j \leq M'$ give s_M) and at $n = M+1$ (where $\gamma(M+1) = 0$ and there are no terms with $j > M$). $\square$

Proposition 5.5 is the rigorous core of the ratio law. Since the weights $\omega_{n,j}$, $j \leq M'$, are concentrated near $j = M'$ and $\omega_{n+1,j}/\omega_{n,j} = p(n+1)/(n+1-j)$, the ratio s_{M+1}/s_M is $p(M+1)/(M+1-M')(1+o(1))$ provided the block $(M'-K, M']$ with $K = O(\log M)$ carries most of s_M ; then the proposition forces

$$M+1 \geq \frac{q}{1-2p} (M' - K)(1 - o(1)), \quad (12)$$

a rigorous *geometric lower bound* on the growth of magic numbers, with the same ratio $q/(1-2p)$ at which the older-record term stops decaying faster than q^n in Lemma 5.3. Note that $q/(1-2p) > 1/q$ always, since $1-2p = 2q-1$ and $q^2 - (2q-1) = (1-q)^2 = p^2 > 0$; so a birth can only occur after the freeze, and $q/(1-2p) < r(p)$ for all p (Table 1); the observed ratios cross $q/(1-2p)$ after a few cycles and continue towards $r(p)$. The block-domination hypothesis is where the proof of (12) is still open; the theorem below shows what it, together with the exact identity, yields for one full cycle, and how the estimates propagate from one cycle to the next.

5.1 The mechanism of a cycle

Conjecture 5.6 (cycle structure). *For every $p \in (0, \frac{1}{2})$ and every k large enough, the magic numbers of Theorem 4.2 satisfy $M_{k+1} + 1 \geq (1+c)qM_k/(1-2p)$ for some fixed $c > 0$, the run of decreases after M_k is bounded, and w increases from its end to the birth of M_{k+1} .*

Two exact facts turn the transition between the memoryless Phase 1 and the next birth into a one-dimensional recursion. Write $P_n := 1 - U_n - V_n = \mathbb{P}(M' < \text{Bin}(n, q) \leq M)$.

Lemma 5.7 (when the binomial tail decays slower than q^n). *Let $\tau_n := \mathbb{P}(\text{Bin}(n, q) \leq M)$ and $n^* := qM/(1-2p)$. Then $\tau_{n+1}/\tau_n = 1 - q\mathbb{P}(\text{Bin}(n, q) = M)/\tau_n$, and with $\kappa_n := Mp/((n+1-M)q)$ the ratio of the two top terms of τ_n ,*

$$\frac{\tau_{n+1}}{\tau_n} \geq q \quad \text{if} \quad \frac{\kappa_n}{1-\kappa_n} \left(1 - O\left(\frac{\log M}{M}\right)\right) \geq \frac{1-2p}{p}, \quad \frac{\tau_{n+1}}{\tau_n} \leq p + q\kappa_n.$$

Since $\kappa_n/(1-\kappa_n) \geq (1-2p)/p$ iff $\kappa_n \geq (1-2p)/q$ iff $n+1 \leq n^*$, the tail τ_n/q^n increases up to $n = n^*(1 - O(\log M/M))$ and decreases geometrically afterwards. In particular $\rho_n^+ = p(n+1)/(n+1-M') \leq \lambda q$ with $\lambda = p(1+c)/(p+cq) < 1$ whenever $n+1 \geq (1+c)qM'/(1-2p)$.

Proof. The identity is $\mathbb{P}(\text{Bin}(n+1, q) \leq M) = \tau_n - q\mathbb{P}(\text{Bin}(n, q) = M)$. Write $\tau_n/\mathbb{P}(\text{Bin}(n, q) = M) = \sum_{i \geq 0} \prod_{l < i} \kappa_n^{(l)}$ with $\kappa_n^{(l)} = (M-l)p/((n+1-M+l)q)$ decreasing in l from κ_n ; truncating at $i = K = O(\log M)$ gives $\geq 1 + \kappa_n/(1-\kappa_n) - O(K/M)$, and $1/(1-\kappa_n)$ is an upper bound; then $\tau_{n+1}/\tau_n \geq q$ iff $\tau_n/\mathbb{P}(\text{Bin} = M) \geq q/p$. The last claim: $p(n+1)/(q(n+1-M'))$ at $n+1 = (1+c)qM'/(1-2p)$ equals $p(1+c)/(p+cq)$ (use $1-2p = 2q-1$), and it decreases in n . $\square$

Lemma 5.8 (the normalised gap). *For $n > M$ put $\varepsilon_n := \sigma_n + (p/q)^n(1-w_{n-1})/w_{n-1}$ and $A_n := \gamma(n)P_n/(q^n w_{n-1})$. Then $d_n \geq 0$ iff $A_n \geq 1 - \varepsilon_n$, and whenever $\theta_n = 0$,*

$$A_{n+1} = \frac{P_{n+1}}{qP_n} \cdot \frac{w_{n-1}}{w_n} \cdot [(1-P_n)A_n + P_n(1-\varepsilon_n)]. \quad (13)$$

Proof. With $\theta_n = 0$, (10) reads $\gamma(n+1) = (1-P_n)\gamma(n) + q^n w_{n-1}(1-\sigma_n) - p^n(1-w_{n-1})$ and $d_n = \gamma(n) - \gamma(n+1) = q^n w_{n-1}[A_n + \varepsilon_n - 1]$. $\square$

Lemma 5.8 says that the increase after a birth persists exactly as long as $A_n \geq 1 - \varepsilon_n$, with ε_n exponentially small after the run, and that A evolves by an explicit recursion whose only n -dependent input is the tail ratio of Lemma 5.7: in the flat range $A_{n+1} \approx 1/q > 1$; through the transition A is multiplied by $\tau_{n+1}/(q\tau_n)$, which is ≥ 1 up to n^* and < 1 beyond. Thus $\log A$ gains $\approx MI(p)$ up to n^* and then loses $\approx MJ(p, n/n^*)$,

$$I(p) := \int_{1/q}^{q/(1-2p)} \log \frac{px}{q(x-1)} dx, \quad J(p, x) := \int_{q/(1-2p)}^{xq/(1-2p)} \log \frac{q(x'-1)}{px'} dx', \quad (14)$$

and the next birth is where A returns to 1: $n_b = x_b n^*$ with $J(p, x_b) = I(p)$. For $p = 0.45$: $I = 0.6178$, $x_b = 2.409$, $n_b/M = x_b q/(1 - 2p) = 13.251$, exactly the root $r(p)$ of Conjecture 4.7 — the ratio law derived from the exact recursion. On the exact trajectories the criterion $d_n \geq 0 \iff A_n \geq 1 - \varepsilon_n$ holds at every n of every cycle; e.g. $p = 0.25$, $M = 292$: $A = 1.52$ at M/q , 14.1 at $n^* = 438$, 1.036 at 471 and 0.900 at the birth 472. What this does *not* give is a proof: to run (13) one needs the run after M to end and σ to decay, which needs the position of M relative to M' — Conjecture 5.6 is exactly the statement that this regularity propagates from cycle to cycle, and the previous version of this section proved that it does, under an explicit growth hypothesis and for M beyond an unspecified $M_0(p)$; we do not reproduce that induction here.

5.2 A non-increasing tail cannot start late

We now argue by contradiction. Suppose Conjecture 1.4 fails: w is non-increasing on $[N, \infty)$, with N minimal, so that $w_{N-1} < w_N$ and N is a record. Let $M' < N$ be the last strict record before N , and use the notation of Lemma 5.1 with M replaced by N : $\gamma(n) = w_N - w_{n-1}$ (now non-decreasing in n), $s_n, \sigma_n = s_n/(q^n w_{n-1})$, $P_n = \mathbb{P}(M' < \text{Bin}(n, q) \leq N)$. Since $d_n \leq 0$ for all $n > N$, (4) gives for all $n > N$

$$q^n w_{n-1} \geq p^n (1 - w_{n-1}) + s_n + P_n \gamma(n) + Q_n, \quad Q_n = \sum_{j=N+1}^{n-1} \omega_{n,j} (w_j - w_{n-1}) \geq 0. \quad (15)$$

Theorem 5.9 (no late tails). *Fix $c > 0$ and put $\lambda_c := p(1+c)/(p+cq) < 1$. There is $N_0(p, c)$ such that a non-increasing tail as above with $N+1 \geq (1+c)qM'/(1-2p)$ has $N \leq N_0(p, c)$.*

Proof. Flatness. Let $n_1 := \max\{n : P_n \geq \frac{1}{2}\}$; by Chernoff, $n_1 \geq (N - C\sqrt{N \log N})/q$. From (15), $P_n \gamma(n) \leq q^n$, and γ is non-decreasing, so $\gamma(n) \leq \gamma(n_1) \leq 2q^{n_1}$ for all $n \leq n_1$.

Tracking. For $N < k \leq n_1$, (15) rewritten reads $0 \leq -d_k = q^k w_{k-1} (1 - \sigma_k) - p^k (1 - w_{k-1}) - P_k \gamma(k) - Q_k$, so $\sigma_k \leq 1$, and $\sum_{k=N+1}^{n_1} (-d_k) = \gamma(n_1 + 1) \leq 2q^{n_1}$. Since $P_k \gamma(k) + Q_k \leq \gamma(k) \leq 2q^{n_1}$, each summand satisfies $q^k w_{k-1} (1 - \sigma_k) \leq 2q^{n_1} (1 + 2n_1) + p^k$, i.e.

$$1 - \eta_k \leq \sigma_k \leq 1, \quad \eta_k := \frac{2(1 + 2n_1)q^{n_1-k} + (p/q)^k}{a_\infty}, \quad N < k \leq n_1.$$

Thus $s_{k+1}/s_k = q(w_k/w_{k-1})\sigma_{k+1}/\sigma_k \geq q(1 - \eta_{k+1})(1 - q^k/a_\infty)$: the older-record term must track $q^k w$ exactly, up to exponentially small errors, on the window $(N, n_1]$.

Decay. By Lemma 5.3, $s_{k+1} \leq \rho_k^+(s_k + |d_k|U'_k)$ with $\rho_k^+ = p(k+1)/(k+1-M')$, and $|d_k|U'_k/s_k \leq q^k U'_k/(q^k w(1 - \eta_k)) \leq 2U'_k/a_\infty$, where $U'_k = \mathbb{P}(\text{Bin}(k, q) \leq M')$ is exponentially small for $k \geq N$ because $Nq \geq (1+c)q^2 M'/(1-2p) > M'$. The hypothesis $N+1 \geq (1+c)qM'/(1-2p)$ gives $\rho_k^+ \leq \lambda_c q$ for $k \geq N$ (the same computation as in Lemma 5.7). Hence $s_{k+1}/s_k \leq \lambda_c q(1 + 2U'_k/a_\infty)$, which contradicts the tracking bound as soon as $\eta_{k+1} + q^k/a_\infty + 2U'_k/a_\infty < 1 - \lambda_c$, i.e. for N large. $\square$

The proof is uniform in p and uses nothing beyond (4). The same three steps give a conditional form of the gap bound that Section 4 needs.

Proposition 5.10 (gap bound, conditional). *Let $M' < M$ be consecutive records and suppose that after the run following M the sequence is non-decreasing, with $w_{n-1} \geq \max_{M < j < n} w_j$, up to the next decrease at n_b (Phase 1 and Phase 2 without intermediate dips), and that $\sigma_k \leq \frac{1}{2}$ for $k \geq k_1$ and $n_b > k_1 + 1$, where $k_1 \geq M/q + C\sqrt{M \log M}$ is such that $\sum_{i > k_1} \mathbb{P}(\text{Bin}(i, q) \leq M) \leq 1$. Then*

$$w_M - W(p) \geq w_M - w_{n_b-1} \geq \frac{a_\infty}{4e^2} q^{k_1}.$$

Proof. With $\theta_n = 0$, (10) gives $\gamma(n+1) = (1 - P_n)\gamma(n) + q^n w_{n-1}(1 - \sigma_n) - p^n(1 - w_{n-1}) \geq V_n \gamma(n) + q^n a_\infty/4$ for $k_1 \leq n < n_b$; iterate and keep the first summand: $\gamma(n_b) \geq \frac{a_\infty}{4} q^{k_1} \prod_{i>k_1} V_i \geq \frac{a_\infty}{4e^2} q^{k_1}$, as $\log(1-x) \geq -2x$ for $x \leq \frac{1}{2}$. Finally $W(p) \leq w_{n_b-1}$ because $n_b - 1$ is a record: the increase phase leaves every later value below w_{n_b-1} by the same computation, since $\tilde{\sigma}$ decays after n_b (Lemma 5.3). $\square$

The hypothesis of Proposition 5.10 is the regularity of the cycle after M , and the only way we know to guarantee it is the growth condition $M+1 \geq (1+c)qM'/(1-2p)$ for the pair before (then σ decays geometrically by Lemma 5.3, the run is bounded, and Lemma 5.8 carries the increase past k_1). So the gap bound for M rests on the position of M relative to M' , which rests on the cycle before — the induction of the previous version of this section. We do not know how to break this chain from below; Theorem 5.9 shows what can be said without it: a counterexample to Conjecture 1.4 would have to be an *early* tail, $N+1 < (1+c)qM'/(1-2p)$.

5.3 Tracking forces dense records

The tracking step of Theorem 5.9 is a statement about non-negative combinations of binomial weights, and it has a sharp converse: a combination $\sum_{j \leq J} c_j \omega_{k,j}$ with $c_j \geq 0$ can behave like Aq^k on a long window of k only if its mass is concentrated, for every k , near a specific moving index.

Lemma 5.11 (tilting). *Let $c_j \geq 0$ ($0 \leq j \leq J$), $S_k := \sum_{j \leq J} c_j \omega_{k,j}$, and $r_{k,j} := \omega_{k+1,j}/\omega_{k,j} = p(k+1)/(k+1-j)$. Let $\mu_k(j) := c_j \omega_{k,j}/S_k$. Then $S_{k+1}/S_k = \mathbb{E}_{\mu_k}[r_{k,\cdot}]$, $\mu_{k+1}(j) = \mu_k(j) r_{k,j}/\mathbb{E}_{\mu_k}[r_{k,\cdot}]$, and*

$$\frac{S_{k+2}}{S_{k+1}} \geq \left(\frac{S_{k+1}}{S_k} + \frac{\text{Var}_{\mu_k}(r_{k,\cdot})}{S_{k+1}/S_k} \right) \left(1 - \frac{J}{(k+1)(k+2-J)} \right).$$

Proof. The first two identities are $\omega_{k+1,j} = \omega_{k,j} r_{k,j}$. Then $\mathbb{E}_{\mu_{k+1}}[r_{k,\cdot}] = \mathbb{E}_{\mu_k}[r_{k,\cdot}^2]/\mathbb{E}_{\mu_k}[r_{k,\cdot}] = m_k + \text{Var}_{\mu_k}(r_{k,\cdot})/m_k$ with $m_k = S_{k+1}/S_k$, and $r_{k+1,j} = r_{k,j}(1 - j/((k+1)(k+2-j))) \geq r_{k,j}(1 - J/((k+1)(k+2-J)))$. $\square$

Proposition 5.12 (concentration). *Suppose $S_k = Aq^k(1 + \eta_k)$ with $|\eta_k| \leq \eta$ for all k in a window $[K_1, K_2]$ with $K_1 \geq J/q$. Then for every $k \in [K_1, K_2 - 1]$,*

$$\text{Var}_{\mu_k}(j) \leq \left(\frac{q}{p} \right)^2 \left(4\eta k^2 + \frac{J}{p} \right) \quad \text{and} \quad \mathbb{E}_{\mu_k}[j] = j^*(k) + O(\eta k + J/k), \quad j^*(k) := (k+1) \left(1 - \frac{p}{q} \right).$$

In words: the measure $c_j \omega_{k,j}$ is concentrated within $O(\sqrt{J} + k\sqrt{\eta})$ of $j^(k)$, and $j^*(k)$ sweeps the interval $[(1-p/q)K_1, (1-p/q)K_2]$ as k runs through the window.*

Proof. Tracking gives $m_k, m_{k+1} \in [q(1-2\eta), q(1+2\eta)]$, so Lemma 5.11 yields $\text{Var}_{\mu_k}(r) \leq m_k(m_{k+1}/(1-\varepsilon_k) - m_k) \leq q^2(4\eta + \varepsilon_k)(1 + O(\eta))$ with $\varepsilon_k = J/((k+1)(k+2-J)) \leq J/(p(k+1)^2)$ for $k \geq J/q$. Since $j \mapsto r_{k,j}$ is increasing with $\partial_j r_{k,j} \geq p/(k+1)$, $\text{Var}(r) \geq (p/(k+1))^2 \text{Var}(j)$ (pair formula for the variance), which is the first claim. For the second, $r_{k,\cdot}$ is convex, so $r_{k,\mathbb{E}j} \leq \mathbb{E}[r] = m_k$ (Jensen) gives $\mathbb{E}j \leq j^*(k)(1 + O(\eta))$; and $\mathbb{E}[r] \leq r_{k,\mathbb{E}j} + \frac{1}{2} \max r'' \text{Var}(j)$ with $r'' \leq 2p(k+1)/(k+1-J)^3$ gives $\mathbb{E}j \geq j^*(k) - O(\eta k + J/k)$, using $r_{k,j} = q \iff j = j^*(k)$. $\square$

Corollary 5.13 (dense records). *In the situation of Theorem 5.9 (a non-increasing tail from N , previous record M'), apply Proposition 5.12 to $c_j = x_j - w$ ($j \leq J = M'$), which is constant on the blocks between consecutive records below M' , on the window $(N, n_1 - C \log N]$, where the tracking error η is exponentially small. Within a block $(M_i, M_{i+1}]$ the weight $\omega_{k,j}$ grows geometrically in j (ratio $\geq q/(q-p)$ near $j^*(k)$), so μ_k restricted to a block is concentrated at its right end, the record*

M_{i+1} . Hence: for every k in the window there is a record within $O(\sqrt{M'})$ of $j^*(k)$, i.e. the records are $O(\sqrt{M'})$ -dense on the interval $[(1 - p/q)(N + 2), (1 - p/q)(n_1 - C \log N)]$, and in particular $(1 - p/q)n_1 \leq M' + O(\sqrt{M'})$, i.e.

$$N \leq \frac{q^2}{1 - 2p} M' (1 + o(1))$$

— a counterexample tail must start within a factor $q^2/(1 - 2p)$ of the previous record, and it must be preceded by a stretch of relative length $\geq 1/q$ on which records are dense, that is, on which w is non-increasing up to fluctuations of size q^n .

Remark 5.14 (descent). Corollary 5.13 replaces the tail on $[N, \infty)$ by a quasi-tail (dense records) on $[(1 - p/q)N, M']$, one scale factor $1 - p/q$ below. If the tracking argument can be run on a quasi-tail — the missing piece is that between dense records w is not exactly non-increasing, so (15) holds only up to the small increases d_n^+ — the descent continues to scales $(1 - p/q)^k N$, and it must stop at scales of order one, where the sequence is known explicitly (w is strictly decreasing on $\{1, \dots, 5\}$ and $\{1, \dots, n_0(p)\}$ but the records there are consecutive, not $O(\sqrt{\cdot})$ -dense relative to their scale in the required sense once $n \geq 2$). This would be a proof of Conjecture 1.4 uniform in p , with no computation and no induction over cycles. We have not carried out the quasi-tail step.

6 Numerical evidence

All values were computed from the exact Bellman recursion (1) in 220-digit (and, as a check, 320-digit) floating-point arithmetic with `mpmath`; the scripts `simulation/magic_numbers.py` and `simulation/magic_birth.py` in the repository [3] reproduce every number below. The identity (4) was verified to 10^{-220} for all $n \leq 700$ and all p in Table 2.

p	magic numbers on $[1, 700]$	decreases ($n \geq 10$)	osc w	osc b	$w_{700} - b_{700}$
0.15	1..271, 313, 378, 457, 554, 673	10–286, 314–325, ...	$2 \cdot 10^{-25}$	$4 \cdot 10^{-23}$	$1 \cdot 10^{-25}$
0.20	1..126, 148, 197, 265, 359, 489, 671	10–133, 149–160, ...	$1 \cdot 10^{-34}$	$3 \cdot 10^{-17}$	$2 \cdot 10^{-18}$
0.25	1..66, 77, 78, 79, 118, 184, 292, 471	10–69, 78–88, 119–122, ...	$2 \cdot 10^{-44}$	$1 \cdot 10^{-13}$	$6 \cdot 10^{-14}$
0.30	1..38, 54, 102, 201, 408	10–42, 55–60, 103–105, ...	$4 \cdot 10^{-52}$	$5 \cdot 10^{-11}$	$8 \cdot 10^{-11}$
0.35	1..20, 27, 28, 68, 187, 538	10–22, 28–33, 69–71, ...	$1 \cdot 10^{-60}$	$3 \cdot 10^{-9}$	$1 \cdot 10^{-8}$
0.40	1..11, 24, 104, 504	10–14, 25–27, 105–107, 505–506	$5 \cdot 10^{-70}$	$7 \cdot 10^{-8}$	$1 \cdot 10^{-6}$
0.42	1..6, 9, 10, 43, 273	10–13, 44–45, 274–275	$2 \cdot 10^{-83}$	$3 \cdot 10^{-7}$	$6 \cdot 10^{-6}$
0.45	1..4, 15, 167	16–17, 168–169	$1 \cdot 10^{-83}$	$9 \cdot 10^{-7}$	$2 \cdot 10^{-4}$
0.47	1..3, 27, 657	28–29, 658–659	$9 \cdot 10^{-89}$	$1 \cdot 10^{-6}$	$2 \cdot 10^{-4}$
0.49	1..3, 152	153–154	$2 \cdot 10^{-92}$	$1 \cdot 10^{-6}$	$2 \cdot 10^{-4}$

Table 2: Magic numbers, decreases, and relative oscillation $\text{osc } x := (\max - \min)/\max$ over $n \in [350, 700]$ of the optimal value w and the greedy value b . “1.. k ” abbreviates the initial run $1, 2, \dots, k$ of consecutive magic numbers, on which greedy is optimal and $w = b$ exactly.

Three features stand out. First, w converges while b does not: the two oscillation columns differ by 10 to 80 orders of magnitude. Second, for small p the greedy strategy is optimal on a long initial stretch ($w_n = b_n$ exactly for $n \leq 66$ at $p = 0.25$, for $n \leq 271$ at $p = 0.15$, for $n \leq 682$ at $p = 0.10$), after which isolated magic numbers appear; this initial run is the regime $\mathcal{M}_p \supseteq [1, n_0(p)]$ in which w is strictly decreasing, and $n_0(p) \rightarrow \infty$ as $p \rightarrow 0$. Third, each birth is followed by a run of two to a dozen consecutive decreases (the transient of Phase 1: the birth is marginal, so the first gap $|d_{M+1}|$ is much smaller than q^{M+1} , and a few more decreases are needed before $\gamma(n) \approx q^n w$ is reached),

after which w increases until the next birth. The size of a decrease relative to the bound (3) is small but not negligible: at $p = 0.45$, $-d_{168}/q^{168} \approx 0.03$.

The next births beyond $n = 700$ can be extrapolated from (8) with $\gamma_M(\infty)$ replaced by $w_M - w_{700}$ (an upper bound, so the prediction is an upper bound): $n \lesssim 2140$ for $p = 0.45$ and $n \lesssim 1843$ for $p = 0.42$.

7 Discussion

The greedy gap. Since $w \geq b$ and w converges while b oscillates, $w_n - b_n$ oscillates with the amplitude of b around a positive mean; at $p = 0.45$ the mean gap is $1.5 \cdot 10^{-4}$, at $p = 0.25$ it is $6 \cdot 10^{-14}$. Greedy is therefore not asymptotically optimal, but its loss is a constant, not a growing quantity. A proof that $W(p) > \limsup_n b_{n,p}$ for all $p < \frac{1}{2}$ would follow from the strict inequality $w_m > b_m$ at the first isolated magic number, combined with the supersolution property of w .

Status of Conjecture 1.4. Suppose w were eventually non-increasing, from M on. Then $K_j^{(n)} = w_j$ for $j \geq M$, so beyond M the value satisfies the *greedy* recursion with modified data, and the analysis of [2] applies: w has a log-periodic profile, which by Theorem 1.1 must be constant. This was the reduction of our earlier approach (its “Gap D”); it asks for non-cancellation, and we now know that cancellation is precisely what the optimal player achieves whenever a magic number is available. What is needed instead is an argument that a non-increasing tail cannot be self-consistent: from (4), $d_n \leq 0$ for all $n > M$ forces $\sum_{j \geq M} \omega_{n,j}(w_j - w_{n-1}) \leq q^n$, i.e. the binomial average of the decrements at scale nq is at most q^n , which after iteration bounds $w_n - W(p)$ by $O(q^{n/q})$; a coupling of the games with n and $n+1$ coins gives complementary bounds. We have not closed this gap. We note that the conjecture is the exact analogue of van Doorn’s “infinitely complex” conjecture [4, §10] for the all-heads objective: the answer to “with two tails, keep two coins or three?” changes infinitely often iff the set of magic numbers is infinite.

The amplitude functional. The conjecture can be made quantitative. Fix a magic number M and let $z = z^{(M)}$ be the *forced greedy continuation*: $z_n = w_n$ for $n \leq M$ and $z_n = \sum_{j=0}^{n-1} \omega_{n,j} x_j$ for $n > M$, with $x_0 = 1$, $x_j = \max_{j \leq m \leq M} w_m$ for $1 \leq j \leq M$ and $x_j = z_j$ for $j > M$. This is what w would be if it were non-increasing from M on, so Conjecture 1.4 says: *no $z^{(M)}$ is eventually non-increasing*. Poissonising, $Z(t) := \mathbb{E}[z_{\text{Po}(t)}]$ satisfies the exact functional equation $Z(t) = (1 - e^{-pt})Z(qt) + S(t)$ with $S(t) = (1 - e^{-pt})e^{-qt}P(qt) + e^{-t}R(t)$, where $P(u) = \sum_{j \leq M} (x_j - w_j)u^j/j!$ and $R(u) = \sum_{n \leq M} r_n u^n/n!$, $r_n = w_n - \sum_{j < n} \omega_{n,j} x_j$, are polynomials determined by the data up to M . Iterating, $Z(t) = \sum_{k \geq 0} \prod_{i < k} (1 - e^{-pq^i t}) S(q^k t)$, and exactly as in [2] one obtains $Z(t) = \Phi_M(\log t) + O(e^{-ct})$ with the L -periodic profile $\Phi_M(\tau) = \sum_{k \in \mathbb{Z}} \Pi(q^k e^\tau) S(q^k e^\tau)$, $\Pi(u) = \prod_{j \geq 1} (1 - e^{-pu/q^j})$, whose Fourier coefficients are

$$A_m(M) = \frac{1}{L} \int_0^\infty u^{2\pi i m/L-1} \Pi(u) S(u) du. \quad (16)$$

If $A_1(M) \neq 0$, the profile is non-constant, and after analytic de-Poissonisation (with error relative to the oscillation, since the kernel is linear) $z^{(M)}$ oscillates and is not eventually monotone: Conjecture 1.4 holds at M . Thus

$$\text{Conjecture 1.4} \iff A_1(M) \neq 0 \text{ for every magic number } M.$$

Formula (16) is explicit and was evaluated numerically (`simulation/continuation_amplitude.py`). Validation: for $M = 0$ (pure greedy) A_0 reproduces the mean of b_n to ten digits and $4|A_1|$ its peak-to-peak oscillation ($p = 0.45$: $3.4 \cdot 10^{-7}$ vs. $3.8 \cdot 10^{-7}$ on $n \in [350, 700]$); for $p = 0.45$, $M = 15$, a direct computation of $z^{(15)}$ up to $n = 3000$ gives mean 0.411292054 and peak-to-peak $3.34 \cdot 10^{-8}$, against $A_0 = 0.411292054$ and $4|A_1| = 3.24 \cdot 10^{-8}$ — and $z^{(15)}$ indeed changes direction 11 times on $[100, 3000]$. At the magic numbers:

p	M	$ A_1(M) $	$ A_2(M) $	$ A_1(M) /q^M$
0.45	15	$8.1 \cdot 10^{-9}$	$8 \cdot 10^{-13}$	$6 \cdot 10^{-5}$
0.45	167	$2.8 \cdot 10^{-47}$	$1 \cdot 10^{-47}$	$6.4 \cdot 10^{-4}$
0.25	79	$1.1 \cdot 10^{-15}$	$3 \cdot 10^{-20}$	$8 \cdot 10^{-6}$
0.25	118	$3.4 \cdot 10^{-19}$	$6 \cdot 10^{-22}$	$1.9 \cdot 10^{-4}$
0.25	184	$3.0 \cdot 10^{-27}$	$5 \cdot 10^{-29}$	$3.0 \cdot 10^{-4}$
0.25	292	$1.1 \cdot 10^{-40}$	$9 \cdot 10^{-42}$	$3.6 \cdot 10^{-4}$
0.25	471	$5.0 \cdot 10^{-63}$	$1 \cdot 10^{-63}$	$3.3 \cdot 10^{-4}$

(The values are stable under raising the working precision from 110 to 150 digits and the quadrature degree.) Two facts emerge. First, $A_1(M) \neq 0$ at every magic number examined, with $|A_1| \gg |A_2|$: the first harmonic dominates and the continuation oscillates. Second, $|A_1(M)|/q^M$ stabilises along the magic numbers: the amplitude lives on the scale q^M of the increments at the birth.

Where the amplitude comes from. The true sequence has the same representation with the infinite data $x_j^\infty := \sup_{m \geq j} w_m$ and $r_n^\infty := w_n - \sum_{j < n} \omega_{n,j} x_j^\infty$, and since w converges its profile is constant: $A_m(\infty) = 0$ for $m \neq 0$ (granted a summable rate for $w_n \rightarrow W(p)$, see below). The data of $z^{(M)}$ agree with the infinite data for indices $\leq M$, so by linearity of (16)

$$A_1(M) = -\frac{1}{L} \int_0^\infty u^{2\pi i/L-1} \Pi(u) T_M(u) du, \quad T_M(u) = (1-e^{-pu})e^{-qu} \sum_{j>M} g_j \frac{(qu)^j}{j!} + e^{-u} \sum_{n>M} r_n^\infty \frac{u^n}{n!}, \quad (17)$$

with $g_j := x_j^\infty - w_j \geq 0$ the gap to the next record and $r_n^\infty = -\sum_{M < j < n} \omega_{n,j} (x_j^\infty - K_j^{(n)}) \leq 0$: the amplitude is a functional of the *future* of the trajectory. Evaluating (17) with the data restricted to a window $(M, J]$ reproduces $A_1(M)$ to all printed digits as soon as $J \geq M/q$ (`simulation/future_data_amplitude.py` e.g. $p = 0.25$, $M = 292$: $J = 300$ gives 76%, $J = 389 = M/q$ gives 100% of $|A_1|$). *The amplitude is generated entirely in Phase 1 of the cycle following the birth*, where the trajectory is governed by the gap recursion (7).

A one-dimensional model, and $\Psi_p(m)$. In Phase 1, $g_j = \gamma(j+1) - \gamma(\infty)$ and $r_n^\infty = -g_{n-1}(1 - \mathbb{P}(\text{Bin}(n, q) \leq M) - q^n)$, where γ solves $\gamma(M+1) = 0$, $\gamma(n+1) = (1 - \mathbb{P}(\text{Bin}(n, q) \leq M))\gamma(n) + q^n W - s_n - p^n(1 - W)$, and s_n is the older-record term. Feeding these model data into (17) reproduces the true $A_1(M)$ to five or six digits when s_n is taken from the data (`simulation/phase1_model_amplitude.py`: $p = 0.25$, $M = 184$: $3.03804 \cdot 10^{-27}$ vs. $3.03827 \cdot 10^{-27}$; $M = 292$ and $p = 0.45$, $M = 167$: all printed digits). The older-record term is itself explicit at leading order, $s_n = m q^{M+1} W \mathbb{P}(\text{Bin}(n, q) \leq M') / \mathbb{P}(\text{Bin}(M+1, q) \leq M')$ with M' the previous magic number, so the model depends on (p, W, M, M') and a single number, the *marginality* $m := s_{M+1} / (q^{M+1} w_M) \in [\rho/q, 1)$ of the birth ($\rho = s_{M+2} / s_{M+1}$; observed $m = 0.92$ – 0.97). Because the model data are affine in s , the resulting $\Psi_p(m) := A_1^{\text{model}} / q^M$ is *affine in m* , so its non-vanishing on the admissible interval is decided by two evaluations. Tabulating (`simulation/psi_of_m.py`):

	$p = 0.25$ ($M = 292$, $M' = 184$)	$p = 0.35$ (538, 187)	$p = 0.45$ (167, 15)
admissible m	$[0.88, 1)$	$[0.82, 1)$	$[0.90, 1)$
$ \Psi_p(0.8) $	$4.6 \cdot 10^{-4}$	$5.9 \cdot 10^{-4}$	$1.0 \cdot 10^{-3}$
$ \Psi_p(1.0) $	$3.0 \cdot 10^{-4}$	$3.7 \cdot 10^{-4}$	$3.9 \cdot 10^{-4}$
$\arg \Psi_p/\pi$ on $[0.8, 1]$	-0.59 to -0.62	-0.22 to -0.23	-0.17 to -0.20

The modulus stays above $3 \cdot 10^{-4}$ and the phase moves by less than 0.04π : $\Psi_p(m)$ is far from zero on the whole admissible range, and — being a one-parameter family of complex numbers — has no reason to vanish anywhere.

Proof architecture for Conjecture 1.4. The above reduces the conjecture to the following chain, of which the first two links are exact and the others are the remaining work.

- (i) *Reduction.* w eventually non-increasing from $M \Rightarrow w = z^{(M)} \Rightarrow \Phi_M$ constant $\Rightarrow A_1(M) = 0$. (Exact: product formula and $Z(t) \rightarrow$ profile.)
- (ii) *Future-data formula* (17). Exact given $A_1(\infty) = 0$, which needs $\sum_k \sup_{n \geq q^{-k}} |w_n - W(p)| < \infty$ — a rate of convergence. Numerically $|w_n - W| \leq Cq^{n/r(p)}$; a proof should come from the cycle structure.
- (iii) *Cycle theory.* After a birth at M : the dip run has length given by the (m, ρ) formula, then w increases through M/q , and the gap follows (7) with the older-record term $s_n = (1 + o(1))mq^{M+1}W\mathbb{P}(\text{Bin}(n, q) \leq M')/\mathbb{P}(\text{Bin}(M+1, q) \leq M')$, all with relative errors $o(1)$ as $M \rightarrow \infty$. This is the substantial part; the identity (4) and the flatness bound $w_M - w_n \leq q^{n+1}/\mathbb{P}(\text{Bin}(n, q) \leq M)$ are its rigorous starting points.
- (iv) *Non-vanishing.* $\Psi_p(m) \neq 0$ for $m \in [\rho/q, 1]$ and M'/M in a neighbourhood of $1/r(p)$: an affine function of m , verifiable by interval arithmetic for each p ; the table above shows the margin is large.
- (v) *Transfer.* $A_1(M) \neq 0 \Rightarrow z^{(M)}$ not eventually monotone: analytic de-Poissonisation with error relative to the oscillation, obtained from the Jacquet–Szpankowski theorem applied to $z_n^{(M)} - A_0(M)$, whose Poisson transform is $O(|A_1|)$ in a cone for $|t| \geq T_0(M)$.
- (vi) *Small M .* For the finitely many magic numbers below the range where (iii) applies, $A_1(M) \neq 0$ by direct rigorous evaluation of (16).

Steps (i), (ii) (modulo the rate), (iv) (numerically) and (v) (standard) are in place; (iii) is where the proof has to be written.

Towards a proof of the ratio law.

Towards a proof of the ratio law. The two-phase analysis uses three approximations: neglect of the older records, the freeze estimate $\gamma_M(\infty) \approx q^{M/q}$, and the binomial-tail asymptotics. The first is harmless (older records enter with weights $\mathbb{P}(\text{Bin}(n, q) \leq m_{K-1})$, smaller by a factor $(p/q)^{M-m_{K-1}}$), the third is standard. The freeze estimate is where an error term would have to be controlled: $\gamma_M(\infty) = \sum_{n > M} S_n q^n w$ with survival products $S_n = \prod_{k=M+1}^{n-1} \mathbb{P}(\text{Bin}(k, q) > M)$, which is a sum dominated by $n \approx M/q$ up to a factor sub-exponential in M . Since only $\log \gamma_M/M$ enters (9), such factors do not affect $r(p)$, and Conjecture 4.7 should be provable along these lines once Conjecture 1.4 is available.

The limit $W(p)$. Theorem 1.1 defines $W(p)$ on all of $(0, 1)$: $a_\infty(p) \leq W(p) \leq p$ for $p \leq \frac{1}{2}$, so $W(p) \rightarrow 0$ as $p \rightarrow 0$; $W(\frac{1}{2}) = \frac{1}{2}$; and $p \leq W(p) < 1$ for $p > \frac{1}{2}$ by [1]. Whether W is continuous or monotone on $(0, \frac{1}{2})$ is open; the structure of magic numbers, which are born and die as p varies, suggests that W is at best piecewise smooth.

Formal verification. Lemma 2.1, Theorem 1.1 and Theorem 1.2 are statements about the recursion (1) alone and use only finite sums and the geometric series; they are natural targets for the Lean 4 formalization of [1], which already contains the definition of w and the bound $w_{n,p} < \frac{1}{2}$ for $p < \frac{1}{2}$.

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